Sahaj Bhadja

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Summary

Software engineer with 2+ years building **C++/Python systems**, test automation, simulation pipelines, and cross-platform software; maintained 20+ unit/integration tests, processed 100K+ records, performed root-cause analysis, and delivered CI/CD-backed production releases.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python services and normalized data systems** for 500+ users, implementing validation, indexed queries, and optimized access patterns that reduced application latency from 1.2 seconds to 350 milliseconds.
- Implemented **reconciliation, error handling, and failure isolation** across 5,000+ monthly jobs, achieving 99%+ reliability while accelerating root-cause diagnosis of malformed inputs, failed integrations, and recurring runtime anomalies.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a **concurrent Python processing system** using multithreading and fault-tolerant execution, processing 1,000+ files per batch and 10,000+ daily requests while maintaining stable behavior under peak production workloads.
- Implemented **secure input validation and deterministic workflows** for PDF, DOCX, and ZIP files, adding persistent results, usage logging, and fallback handling to reduce processing turnaround 60% and improve diagnosability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python simulation and evaluation pipelines** processing 100K+ complex trajectory records, creating reproducible test scenarios and benchmark comparisons that improved long-horizon planning performance 18% across controlled experiments.
- Analyzed **model behavior, performance constraints, and failure patterns** across three architectures, documenting evidence and technical findings that guided software improvements and increased closed-loop system stability 15% across simulation tests.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed **object-oriented software components** using Singleton and Factory patterns, API integrations, SQL, and caching, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability across distributed workflows.
- Automated **cross-platform builds and 20+ unit/integration tests** with GitHub, Jenkins, PyTest, and Mockito, generating coverage summaries, CSV test reports, and failure logs across Windows, Linux, and macOS releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **modular software across eight microservices** for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through REST APIs, WebSockets, and Azure SQL while reducing sync conflicts 33%.
- Implemented **unit, integration, and end-to-end testing** with Jest, Selenium, Snyk, and dependency checks, then automated Docker deployments through Jenkins CI/CD, cutting release time 50% across staged environments.

Skills

Languages: C++, Python, Java, SQL, Bash, JavaScript, TypeScript
Test Engineering: Unit testing, integration testing, end-to-end testing, PyTest, Mockito, Jest, Selenium
Software Design: OOP, Singleton, Factory, data structures, algorithms, multithreading, simulation, root-cause analysis
Systems & Data: Linux, Windows, macOS, PostgreSQL, MySQL, SQL Server, REST APIs, data validation
Delivery: Git, GitHub, Jenkins, GitHub Actions, CI/CD, Docker, Agile/Scrum, test reports, failure logging

Projects

CPU Scheduling Simulator Live | *C++, React.js, Algorithms*
- Implemented **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and dynamic performance metrics, testing 1,000+ process combinations while reducing average turnaround time 25% across benchmarks.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a **containerized Python processing application** on AWS ECS/Fargate with CodePipeline, automated validation checks, and CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.

HMIS Live | *TypeScript, MERN, REST APIs, Sentry*
- Built a **secure monitored application** with role-based workflows, REST integrations, TypeScript dashboards, notifications, and Sentry diagnostics, reducing record retrieval time 2x and administrative workload 30% while improving defect visibility.